The iliac crest is an unreliable level landmark in transitional lumbosacral anatomy: a morphometric study of 802 screening CT examinations

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Abstract

Introduction. Surgeons locate the lumbar spine by palpating the iliac crest, which is taught to lie at the L4–5 interspace. Lumbosacral transitional vertebrae make level assignment ambiguous by altering the vertebral count. Whether the palpable landmark remains reliable in those same patients has not been quantified.

Methods. 802 abdominal CT examinations from a colorectal cancer screening cohort were segmented for every vertebra, rib, hip and femur. 738 are recorded female (393) or male (345); 11 carry DICOM's *other* value and 53 no sex field. Age is present for 709 (median 59, range 50–89). For each case we recorded the vertebral level reached by the highest point of the iliac crest, and the number of rib-free vertebrae between the lowest rib and the sacrum — a count-free description of transitional status that does not require naming a level.

Results. With the typical five rib-free vertebrae ($n = 706$) the crest reached L4 in 87.1% of cases (95% CI 84.4–89.4). With four rib-free vertebrae ($n = 28$) it reached L4 in 46.4% (29.5–64.2), reaching L5 in 28.6% (15.3–47.1) and L3 in 25.0% (12.7–43.4). With six ($n = 25$) it reached L4 in 64.0% (44.5–79.8). The intervals for the typical and four-vertebra groups do not overlap. Four rib-free vertebrae arises either from a lumbar rib or from a sacralised lowest segment; the two routes gave the same L4 rate (46.7% vs 46.2%) but opposite directions of error, the crest appearing caudal in 7 of 8 misplaced cases with a lumbar rib and cranial in 6 of 7 without (Fisher exact $p = 0.010$).

Conclusions. The iliac crest indicates the L4–5 interspace in roughly seven of eight patients with typical anatomy and in fewer than half of those with four rib-free vertebrae, in whom it is as likely to point at L5 as at L3. The two sources of level error do not cancel: the

landmark is least trustworthy in precisely the patients in whom the count is already uncertain. Intraoperative imaging confirmation should not be waived on the basis of a palpable crest when transitional anatomy is suspected. The direction of the error is not random: it depends on which end of the vertebral count moved, which is itself visible on preoperative imaging.

1 Introduction

Level identification in the lumbar spine rests on two things: a count, and a palpable landmark. Both are known to be imperfect. What has not been established is whether their errors are independent — and if they are not, a surgeon resolving an ambiguous count by palpating the crest is relying on the wrong thing at the wrong moment.

2 Materials and Methods

2.1 Cohort

Records derive from the publicly available CTSpinoPelvic1K collection, built from the TCIA CT COLONOGRAPHY imaging archive with vertebral and pelvic segmentations unified under a single label scheme. All examinations were performed for colorectal cancer screening; no participant was imaged for a spinal indication, and the lower age bound of 50 reflects the screening guideline rather than any selection applied here.

Ethical approval: the source collections are de-identified and publicly released; [VERIFY institutional determination].

2.2 Measurements

All measurements are automated and deterministic; the code is public [repository URL].

Iliac crest level. The most superior point of either iliac crest was located, and the vertebral level whose segmentation spans that height was recorded.

Transitional status, without naming a level. We counted the vertebrae lying between the lowest rib-bearing vertebra and the sacrum. This is an interval count rather than a level assignment, and it remains valid without deciding whether a given segment is a sacralised L5 or a lumbarised S1 — a distinction a spine-limited field of view cannot make. Five is typical; four and six indicate transitional anatomy.

A count of four can arise at either end of the interval, and the two are recorded separately: a rib borne by the first lumbar-type vertebra shortens the interval from above, while a lowest lumbar segment incorporated into the sacrum shortens it from below. Rib presence is taken from the segmentation directly and requires no level name.

Quality control. Every case passed geometric invariants: label geometry agreeing with its CT in shape, affine and voxel spacing; no identifier outside the published scheme; and left- and right-sided structures on the correct sides of the midline. Rib numbering was verified by rib–vertebra incidence. Of 11,548 ribs, 5,749 have the anchoring vertebra within the field of view and are therefore evaluable; two of those remain offset (0.035%), both field-of-view truncations.

2.3 Analysis

Proportions are reported with Wilson score intervals, which remain within the unit interval at the small counts present in the transitional subgroups — a property the normal approximation does not have.

One test is reported: Fisher's exact test comparing the direction of crest misplacement between the two routes to a count of four (Table 2). It is exact rather than asymptotic because the cell counts are single digits, where the χ^2 approximation is not defensible. No test is reported for the primary comparison in Table 1; the intervals there do not overlap and a p-value would add nothing to that.

3 Results

Table 1: Vertebral level reached by the iliac crest, by rib-free vertebral count. Percentages with 95% Wilson score intervals.

Rib-free count	n	Crest at L4	Crest at L3	Crest at L5
5 (typical)	706	87.1 (84.4–89.4)	8.9 (7.0–11.3)	4.0 (2.8–5.7)
4	28	46.4 (29.5–64.2)	25.0 (12.7–43.4)	28.6 (15.3–47.1)
6	25	64.0 (44.5–79.8)	12.0 (4.2–30.0)	24.0 (11.5–43.4)

Cases in which the crest reached a thoracic level are excluded from Table 1: in those examinations the pelvis was barely within the field of view, which is a fact about the scan rather than about the patient.

3.1 Four rib-free vertebrae is reached by two routes, and they fail in opposite directions

Pooled, the four-vertebra group appears to move the crest in both directions in roughly equal proportion. That apparent symmetry is an artefact of pooling. Four rib-free vertebrae arises two ways: a rib on the first lumbar-type vertebra moves the *upper* boundary of the count caudally, or a lowest lumbar segment incorporated into the sacrum moves the *lower* boundary cranially. Separating them (Table 2) leaves the collapse in reliability unchanged — 46.7% versus 46.2% at L4 — while the direction of the error reverses.

Table 2: Direction of crest misplacement among cases with four rib-free vertebrae, by the route that produced the count. Fisher exact test on the 15 misplaced cases, $p = 0.010$.

Route to four	n	Crest at L4	Crest cranial (L3)	Crest caudal (L5)
Lumbar rib present	15	7 (46.7%)	1	7
No lumbar rib	13	6 (46.2%)	6	1

Where a lumbar rib is present the crest appears caudal to its expected level in 7 of 8 misplaced cases; where the count is four without a lumbar rib it appears cranial in 6 of 7. The mechanism accounts for the sign in both: the crest sits at a fixed skeletal height, and moving either end of the count moves the level *name* assigned to that height in the corresponding direction.

This is an observation rather than a prespecified hypothesis — the split was found in the data and the mechanistic account constructed afterwards — and it rests on 15 misplaced cases. It requires replication before it is relied upon. Its practical content, if it replicates, is that the presence of a lumbar rib is visible on the same preoperative imaging that establishes the count, and it predicts *which way* the crest is wrong rather than merely that it may be.

Where there are six rib-free vertebrae the crest moves predominantly caudally. All three transitional groups are small and no proportion should be read as a precise estimate — the intervals are wide for that reason. What is robust is the collapse in reliability itself: the interval for the typical group does not approach that of the four-vertebra group.

4 Discussion

The clinical reading is a single sentence: *the landmark is least reliable in exactly the patients whose count is already uncertain.* In typical anatomy the crest identifies L4 in roughly seven of eight patients; with four rib-free vertebrae it does so in fewer than half, and it is as likely to point at L5 as at L3. Level-identification error has two sources, and a surgeon who resolves an ambiguous count by palpating the crest is compounding them rather than cross-checking one against the other.

4.1 Limitations

The cohort is asymptomatic and undergoing screening, so these distributions sit closer to a reference range than a surgical series would.

Crest level was determined from supine CT, whereas palpation is performed through soft tissue with the patient prone or lateral. That difference runs in one direction only: palpation cannot be more accurate than the bone it is feeling for, so the figures here are an *upper bound* on landmark reliability rather than an estimate of it. The true reliability of a palpated crest in transitional anatomy is therefore at best what Table 1 reports, and plausibly worse.

Transitional status is described by rib-free count rather than by Castellvi grade. The two describe different things — one a count, the other a morphology — and are not interchangeable. The transitional subgroups are small ($n = 28$ and $n = 25$), which is why the intervals are reported prominently and why no precise prevalence should be read from them.

Data availability

All measurements, the segmentations they derive from, and the code that produced them are publicly available [DOI].

Conflicts of interest

The authors declare no conflicts of interest.

References

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